

Managing the risks of energy industry

Lecture 4

Peter Tankov

ENSAE ParisTech

Outline

- ① Estimation of the solar potential
- ② Electricity storage
- ③ Probabilistic forecasting for renewable energy

Solar power

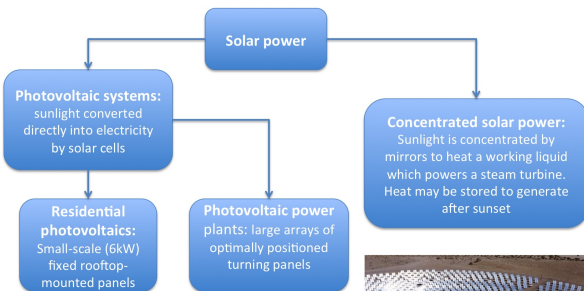


Photo: greentechmedia.com



Photo: new.abb.com



Photo: Sandia National Laboratories/ seia.org

Saudi Arabia wants to host the world's cheapest data centres

With plentiful land and electricity on hand, the kingdom thinks it has found an edge

[Save](#) [Share](#) [Summary](#)



Solar and the city PHOTOGRAPHY: ESA

Dec 17th 2025 | RYANAL | 5 min read

Listen to this story

▶ 0:00 / 0:00

TWO HOURS south of Jeddah, on Saudi Arabia's Red Sea coast, the Al Shuaiba solar farm blankets 50 square kilometres of desert. The first phase of the project, started in 2024, produces 600 megawatts of electricity at just 3.9 Saudi halalas (just over a cent) per kilowatt-hour, nearly a twentieth of the cost of generation at Britain's planned Hinkley Point C nuclear power plant. Saudi Arabia's plan for all this cheap electricity is to power enormous data centres for artificial intelligence (AI).

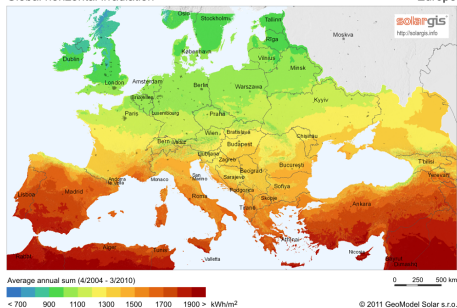
Estimating solar potential

- Solar power potential is almost unlimited: recovering the energy received by just 0.01% of the Earth surface would be enough to supply all energy needs of humanity.
- The power output of a solar cell depends on the insolation and the conversion efficiency
- The insolation depends on the amount of solar energy received by the atmosphere, the absorption in the atmosphere and the **cloud cover**

- The conversion efficiency of commercially available solar cells under standard conditions is around 20%–25%

Global horizontal irradiation

Europe



Estimation of solar potential: top of the atmosphere

- The irradiation of the upper atmosphere can be estimated without error
- Top of the atmosphere irradiance normal to the sun's rays:

$$G_{TOA} = G_{sc} E_0(n), \quad E_0(n) = \left(\frac{r_0}{r(n)} \right)^2, \quad E_0(n) = 1 + 0.033 \cos\left(\frac{2\pi n}{365}\right)$$

$G_{sc} \approx 1361 \text{ W/m}^2$: solar constant, n : day of the year, $r(n)$ actual Earth-Sun distance, r_0 : mean distance (1 AU)

- Horizontal irradiance: multiply by cosine of solar zenith angle θ_z :

$$G_{TOA,h} = G_{sc} E_0(n) \cos \theta_z,$$

- From standard geometry,

$$\cos \theta_z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega, \quad \delta \approx 23.44^\circ \sin(2\pi(n - 81)/365),$$

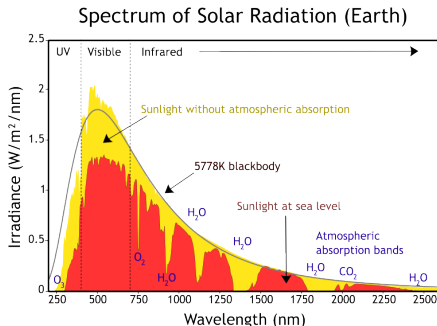
where ϕ is latitude, δ solar declination (angle between sun rays and equator plane), ω hour angle (0 at solar noon, add 15° per hour).

Estimation of solar potential: clear-sky sea-level

- ① Compute the solar zenith angle and the TOA irradiance.
- ② Determine the atmospheric path length (air mass).
- ③ Account for atmospheric attenuation:

- molecular scattering,
- absorption by gases (ozone, water vapor) and aerosols.

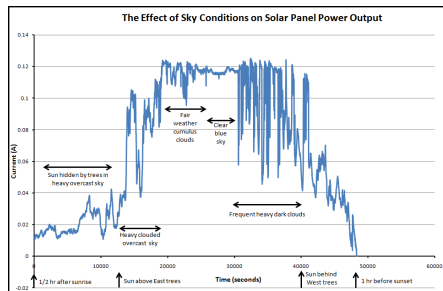
- ④ Obtain direct normal irradiance (DNI) at the surface.
- ⑤ Project DNI onto the horizontal plane and add diffuse radiation to obtain global horizontal irradiance (GHI) $\Rightarrow$ deterministic quantity which can be precisely estimated



Source: Wikipedia (Robert A. Rohde)

Estimating solar potential: cloud cover

- Cloud cover is intermittent but less location-dependent than wind; precise maps of average cloud cover and insolation are available from measurements or satellite data.
- Cloud cover is strongly related to climate change and may present pluriannual variability

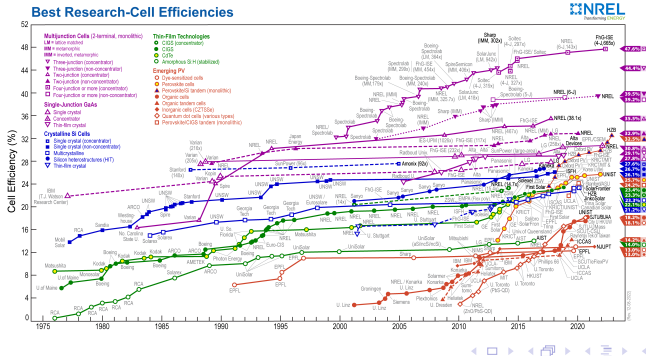


Source: vernier.com

Characteristics of solar panels

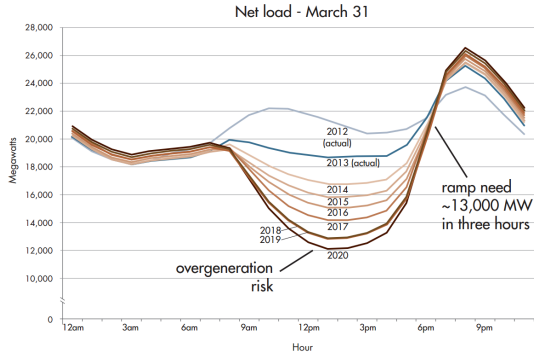
2 types of solar panels: positioned at a fixed optimal angle, or equipped with tracker to be always perpendicular to incoming sun rays.

The power generated by the cell is approximately proportional to the incoming irradiation of the cell



Network integration of solar energy: the duck chart

Ramping requirements of solar generation are more predictable and can be met via storage or demand dispatch



Solar generation profile in California. Source: CAISO, 2013

Illustration for solar potential estimation

- Goal: evaluate the LCOE and the revenue of a small rooftop installation in the south of France.
- The standard capacity is 3kW; the feed-in tariff for this type of installation is 23 cents per kWh.
- The CAPEX is around 9000 Euros and the OPEX ≈ 10 EUR/kW/year .
- The solar power output is downloaded from `renewables.ninja` for years 2014-2020.

Jupyter notebook for the exercise Data file

Another convenient calculator is available here:

https://re.jrc.ec.europa.eu/pvg_tools/en/tools.html

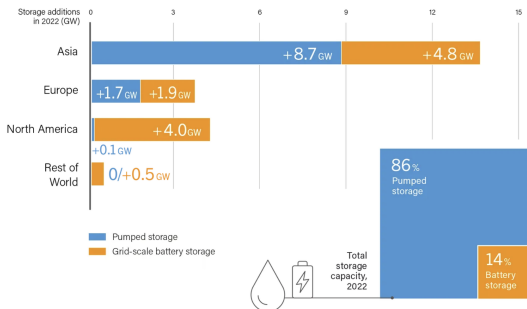
Outline

- ① Estimation of the solar potential
- ② Electricity storage
- ③ Probabilistic forecasting for renewable energy

Introduction

- Large volumes of electricity remain difficult to store, but role of storage is increasing as intermittent renewables are replacing controllable conventional production.
- Pumped storage remains the most available technology (limited by relatively low pumping rate, $\approx 75\%$), but battery storage is starting to catch up.

Capacity Additions of Pumped Storage and Utility-Scale Battery Storage, by Region, and Total Storage Capacity, 2022



Graph source: REN21 2023 global status report

Storage technologies

Technologies differ by their discharge duration and capacity, but also efficiency and cost.

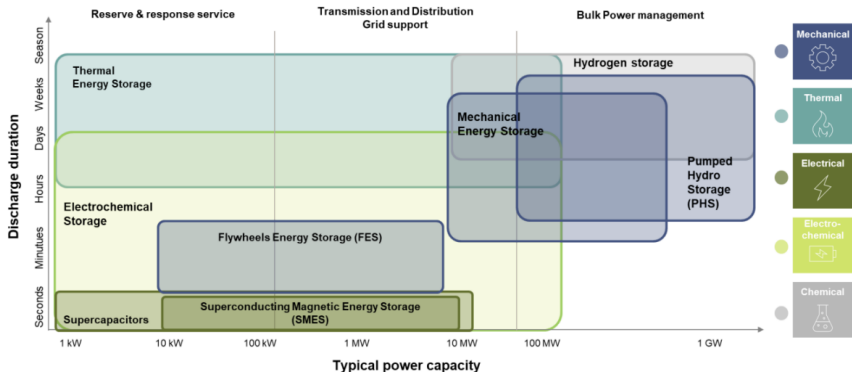
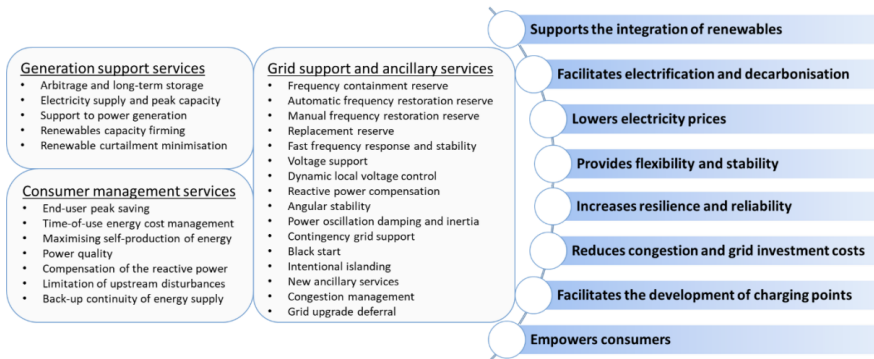


Figure 3. Power ranges and discharge duration of different energy-storage technologies. Source: EASE, Energy Storage Targets 2030 and 2050: Ensuring Europe's Energy Security in a Renewable Energy System, 2022.

Source: European Commission (2023)

Storage services to the electricity system



Source: European Commission (2023)

Suitable technologies per usage type

Table 1. Qualitative Description of Electricity Storage Applications and Technology Suitability

Role	Application	Description	Pumped Hydro	Compressed Air	Flywheel	Lithium Ion	Sodium Sulfur	Lead Acid	Vanadium Redox Flow	Hydrogen	Supercapacitor
System Operation	1. Energy arbitrage	Purchase power in low-price and sell in high-price periods on wholesale or retail market	✓	✓		✓	✓	✓	✓	✓	
	2. Primary response	Correct continuous and sudden frequency and voltage changes across the network			✓	✓	✓	✓	✓	✓	✓
	3. Secondary response	Correct anticipated and unexpected imbalances between load and generation	✓	✓	✓	✓	✓	✓	✓	✓	✓
	4. Tertiary response	Replace primary and secondary response during prolonged system stress	✓	✓		✓	✓	✓	✓	✓	
	5. Peaker replacement	Ensure availability of sufficient generation capacity during peak demand periods	✓	✓		✓	✓	✓	✓	✓	
	6. Black start	Restore power plant operations after network outage without external power supply	✓	✓	✓	✓	✓	✓	✓	✓	✓
	7. Seasonal storage	Compensate long-term supply disruption or seasonal variability in supply and demand	✓	✓				✓	✓		
Network Operation	8. T&D investment deferral	Defer network infrastructure upgrades caused by peak power flow exceeding existing capacity	✓	✓		✓	✓	✓	✓	✓	
	9. Congestion management	Avoid re-dispatch and local price differences due to risk of overloading existing infrastructure	✓	✓		✓	✓	✓	✓	✓	
Consumption	10. Bill management	Optimise power purchase, minimize demand charges and maximise PV self-consumption				✓	✓	✓	✓	✓	
	11. Power quality	Protect on-site load against short-duration power loss or variations in voltage or frequency			✓	✓	✓	✓	✓	✓	✓
	12. Power reliability	Cover temporal lack of variable supply and provide power during blackouts				✓	✓	✓	✓	✓	

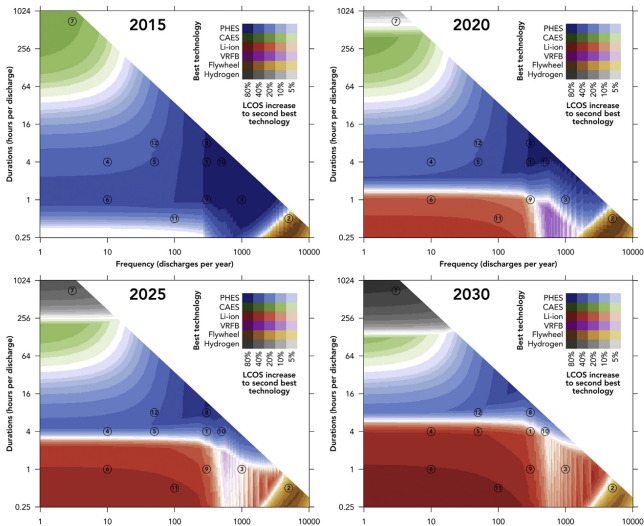
Source: Schmidt et al., Joule 3,81–100, 2019.

Levelized cost of storage

To compare different technologies (for a given application), the following definition may be used:

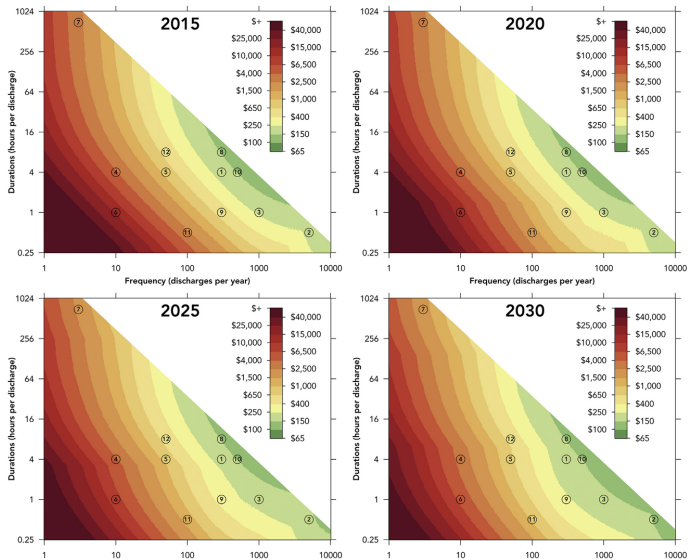
$$\text{LCOS} \left(\frac{\text{USD}}{\text{MWh}} \right) = \frac{\text{Invest. cost} + \sum_{n=1}^N \frac{\text{O\&M cost}}{(1+r)^n} + \sum_{n=1}^N \frac{\text{Charg cost}}{(1+r)^n} + \frac{\text{End-of-life cost}}{(1+r)^{N+1}}}{\sum_{n=1}^N \frac{\text{Elec discharged}}{(1+r)^n}}$$

Most efficient storage technologies per usage type



Source of this and following graph: Schmidt et al., Joule 3,81–100, 2019.

LCOE of the most efficient technology



Example: using battery storage for day-ahead arbitrage

Assume that the owner of a battery only aims to use it for financial arbitrage on the day-ahead market, then we may assume price, denoted by P_t for hour t is deterministic and solve the following problem:

$$\min_{Q_0, q_t, t=1, \dots, 24} \sum_{t=1}^{24} q_t P_t$$

subject to

$$q^{min} \leq q_t \leq q^{max}, \quad Q^{min} \leq Q_t \leq Q^{max}$$

$$\text{and } \sum_{t=1}^{24} q_t = 0, \quad \text{where } Q_t = Q_0 + \sum_{s=1}^t Q_s$$

This linear quadratic problem is easily solved with any optimization software.

Illustration: no price impact

Battery storage capacity:

5MWh

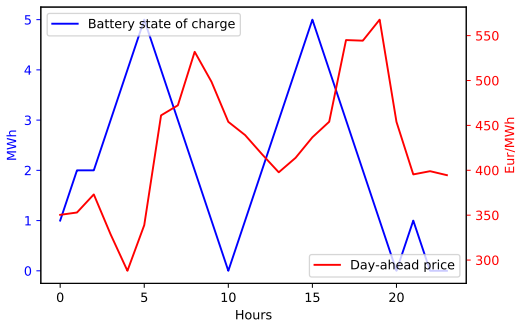
Charge / discharge power:

1MW.

Battery does not impact the market price.

Price trajectory: French day-ahead price of Dec 7, 2022.

Daily revenue: 1222 Eur.



Impact of storage on electricity price

- As more storage enters the market, activity of arbitrageurs will 'shave' the price peaks and fill the troughs, reducing the profits of storage agents.
- The price P_t for every hour t is determined by balancing supply $S_t(P)$ and demand $D_t(P)$ curves:

$$P_t = \min\{P : S_t(P) \geq D_t(P)\}$$

- Assume that near P_t , excess supply is linear:

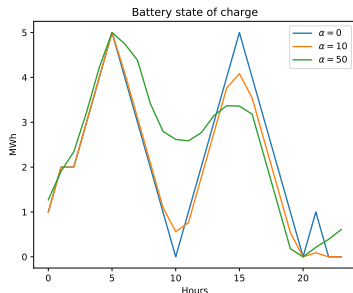
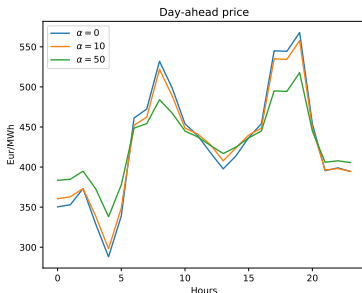
$$S_t(P) - D_t(P) = \alpha^{-1}(P - P_t)$$

- If the battery operator buys quantity q_t for hour t , the price is

$$\hat{P}_t = P_t + \alpha q_t$$

Impact of storage on electricity price

With $\alpha = 10$ EUR/MWh, the daily profit is reduced to 1036 Eur and with $\alpha = 50$ to 593 Eur.



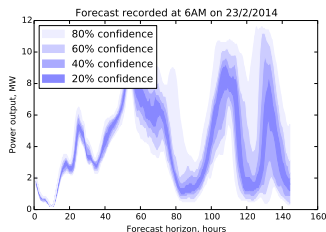
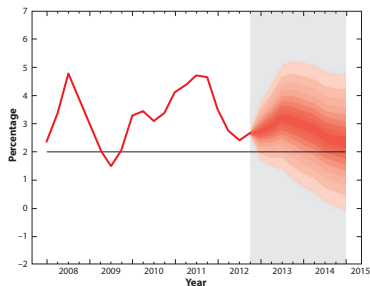
Outline

- ① Estimation of the solar potential
- ② Electricity storage
- ③ Probabilistic forecasting for renewable energy

Introduction

- Probabilistic forecasts serve to quantify the uncertainty in a prediction, and they are an essential ingredient of optimal decision making.
- A probabilistic forecast is an estimate of the **conditional distribution** of the forecast quantity given the available information.
- Usually provided as a set of quantiles, as confidence intervals, or as full predictive distributions (parametric or non-parametric)

Top: forecast of UK inflation from the Bank of England. Bottom: wind power output forecast. Data source: Maïa Eolis.



Motivating example: Bidding strategies for a wind power producer

- Consider a wind power producer aiming to sell the output of the given hour of the next day P in the day-ahead market.
- We assume the producer does not have access to intraday market and pays / receives the adjustment market price for production mismatch.
- Let S be the day-ahead market price and assume that the adjustment market prices are known in advance and equal to $\bar{S} > S$ (bid) and $\underline{S} < S$ (ask).
- The expected profit of the power producer selling π units is

$$\pi S + \underline{S} \mathbb{E}[(P - \pi)^+] - \bar{S} \mathbb{E}[(\pi - P)^+]$$

Computing the optimal bidding strategy

The first order condition for the optimal bid writes:

$$S - \underline{S}\mathbb{P}[P > \pi] - \overline{S}\mathbb{P}[P \leq \pi] = 0$$

The optimal bid is a **quantile** of the distribution of P :

$$\pi = F^{-1}\left(\frac{S - \underline{S}}{\overline{S} - \underline{S}}\right),$$

where F is the distribution function of P .

⇒ optimal bidding in power markets calls for **probabilistic forecasts**

A simplified setting: binary forecasts

- Consider first the simple case when we want to forecast a random variable $Y \in \{0, 1\}$.
- The probabilistic forecast of Y is a probability $p \in [0, 1]$ of the realisation of Y .
- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with elements (p, Y) , where $Y \in \{0, 1\}$ and $p \in \mathbb{R}$.
- Let $\mathcal{A}$ be a subfield of $\mathcal{F}$ (the information set available to forecaster) and assume that p is $\mathcal{A}$ -measurable.
- p is an **ideal forecast** of Y given $\mathcal{A}$ if $p = \mathbb{P}(Y = 1|\mathcal{A}) = \mathbb{E}[Y|\mathcal{A}]$.

Evaluating binary forecasts

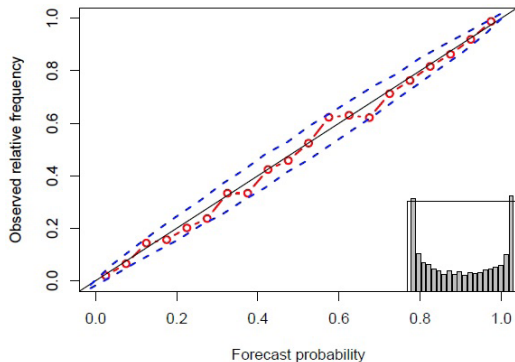
- The forecast p is said to be **calibrated** if

$$\mathbb{E}[Y|p] = p.$$

- An ideal forecast is always calibrated.
- The calibration of a binary forecast may be assessed with a **calibration curve**, e.g., an empirical estimator of $\mathbb{E}[Y|p]$.
- The resolution or **sharpness** of forecasts refers to its capacity of producing definite answers, and can be evaluated from the **distribution of forecast values**.
- A numerical measure of both calibration and sharpness: the **Brier score**

$$B = \mathbb{E}[(p - Y)^2].$$

Evaluating binary forecasts



Example of a calibration curve and forecast value distribution. Blue dashed-curve represents the 95% confidence interval. U-shaped distribution indicates precise forecast. Source: Ranjan and Gneiting (2008).

Continuous forecasts

- We now consider the forecasting of a real-valued random variable Y with continuous distribution (e.g., wind speed)
- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with elements (F, Y) , where Y is real-valued and F is a probability distribution function on $\mathbb{R}$.
- Let $\mathcal{A}$ be a subfield of $\mathcal{F}$ (the information set available to forecaster) and assume that F is $\mathcal{A}$ -measurable.
- F is an **ideal forecast** of Y given $\mathcal{A}$ if $F = \mathcal{L}(Y|\mathcal{A})$.
- Probabilistic forecasts are evaluated in terms of their **calibration** and **sharpness**.

Calibration of probabilistic forecasts

The notion of calibration for binary forecasts, i.e., $\text{Law}(Y|F) = F$ is not suitable in the continuous framework. Instead, one uses several weaker notions.

The random variable

$$Z_F = F(Y)$$

is called the **probability integral transform** of Y .

- The forecast F is **probabilistically calibrated** if $Z_F \sim \mathcal{U}([0, 1])$
 $\Rightarrow \mathbb{P}[Y \leq F^{-1}(u)] = u$, quantiles are evaluated correctly on average
- The forecast F is **marginally calibrated** if $\mathbb{E}[F(y)] = \mathbb{P}[Y \leq y]$ for all $y \in \mathbb{R}$
 $\Rightarrow$ distribution function is evaluated correctly on average

Calibration of probabilistic forecasts

An ideal (or calibrated) forecast is both marginally and probabilistically calibrated but the two properties are not equivalent

If $F(y) = \mathbb{P}[Y \leq y|\mathcal{A}]$ then

$$\mathbb{E}[F(y)] = \mathbb{E}[\mathbb{P}[Y \leq y|\mathcal{A}]] = \mathbb{P}[Y \leq y]$$

$$\mathbb{P}[Z_F \leq z] = \mathbb{E}[\mathbb{P}[Z_F \leq z|\mathcal{A}]] = \mathbb{E}[\mathbb{P}[F(Y) \leq z|\mathcal{A}]] = z.$$

Examples of calibration

Let $Y|\mu \sim N(\mu, 1)$, $\mu \sim N(0, 1)$ so that $Y \sim N(0, 2)$

Let $\tau = -1$ or 1 with probability $\frac{1}{2}$ independently of Y and μ

Forecast	Predictive distribution	Marginally calibrated	Probabilistically calibrated	Ideal
Perfect	$F_1 = N(\mu, 1)$	yes	yes	yes
Climatological	$F_2 = N(0, 2)$	yes	yes	yes
Unfocused	$F_3 = \frac{1}{2}(N(\mu, 1) + N(\mu + \tau, 1))$	no	yes	no
Sign-reversed	$F_4 = N(-\mu, 1)$	yes	no	no

Examples of calibration

Denote by Φ the standard normal CDF.

- Perfect forecast: $F_1 = \mathcal{L}(Y|\sigma(\mu))$
- Climatological forecast: $F_2 = \mathcal{L}(Y)$
- For the unfocused forecaster, let $\Phi_{\pm}(z) = \frac{1}{2}(\Phi(z) + \Phi(z \mp 1))$ and $Z \sim N(0, 1)$. Then,

$$\begin{aligned}\mathbb{P}[F_3(Y) \leq u] &= \frac{1}{2}\{\mathbb{P}[\Phi_+(Z) \leq u] + \mathbb{P}[\Phi_-(Z) \leq u]\} \\ &= \frac{1}{2}\{\mathbb{P}[\Phi_+(Z) \leq u] + \mathbb{P}[\Phi_+(Z + 1) \leq u]\} = u.\end{aligned}$$

- For the sign-reversed forecaster,

$$\mathbb{E}[F_4(y)] = \mathbb{E}[\Phi(y + \mu)] = \mathbb{E}[\mathbb{P}[Z \leq y + \mu|\mu]] = \mathbb{P}[Z \leq y + \mu] = \mathbb{P}[Y \leq y].$$

Sequential interpretation and testing

Assume that we have a sequence of realizations $(F_t, Y_t)_{t=1}^T$.

- To test marginal calibration, we compare the climatological distribution

$$\hat{G}(y) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbf{1}_{Y_t \leq y}$$

with the long-term average of the forecast distribution

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T F_t(y)$$

- To test probabilistic calibration, we test the uniform distribution of the sample $(F_t(Y_t))_{t=1}^T$.
- Formal statistical tests may be used (e.g., Kolmogorov-Smirnov) but attention to possible sequential dependence in the PIT values.

Sharpness of probabilistic forecasts

- Calibration is a joint property of predictions and observations and refers to their statistical consistency;
- Calibration does not distinguish between two ideal forecasts with respect to different information sets;
- Sharpness is a property of forecasts only and refers to the width of the predictive distribution;
- A good probabilistic forecast maximizes sharpness subject to calibration.

Forecast	$\text{Var } F \mathcal{A}$	Marginally calibrated	Probabilistically calibrated	Ideal
Perfect	1	yes	yes	yes
Climatological	2	yes	yes	yes
Unfocused	$\frac{3}{2}$	no	yes	no
Sign-reversed	1	yes	no	no

Example: continuous ranked probability score

CRPS is a **scoring rule** which allows joint assessment of calibration and sharpness

$$crps(F, y) = \int_{-\infty}^{\infty} \{F(x) - \mathbf{1}_{x \geq y}\}^2 dx = \int_{-\infty}^y F(x)^2 dx + \int_y^{\infty} \{F(x) - 1\}^2 dx$$

Finite-sample version:

$$CRPS = \frac{1}{T} \sum_{t=1}^T crps(F_t, Y_t) = \frac{1}{T} \sum_{t=1}^T \int_{-\infty}^{\infty} \{F_t(x) - \mathbf{1}_{x \geq Y_t}\}^2 dx$$

Methods for probabilistic forecasting: quantile regression

Quantile function:

$$Q(\tau) = \inf\{y : F(y) \geq \tau\}, \quad F(y) = \mathbb{P}[Y \leq y].$$

The quantile function is the smallest minimizer of

$$\xi \mapsto \mathbb{E}[\rho_\tau(Y - \xi)],$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{u < 0})$ is the **pinball loss** function.

Sample quantile of a random sample (times series) $\{Y_1, \dots, Y_T\}$: the smallest minimizer of

$$\sum_{t=1}^T \rho_\tau(Y_t - \xi)$$

Quantile regression

Quantile regression with features $(X_t)_{t=1}^T$ (possibly lagged values of Y_t):

$$\arg \min_{\beta \in \mathbb{R}^p} \sum_{t=1}^T \rho_{\tau}(Y_t - X_t^{\top} \beta).$$

The estimated quantile: $\hat{Q}_t(\tau) = X_t^{\top} \hat{\beta}$.

Probabilistic forecasting: estimate each quantile independently, set

$$F_t(y) = \sum_{i=1}^n \frac{i}{n} \mathbf{1}_{\hat{Q}_t(\frac{i}{n+1}) < y \leq \hat{Q}_t(\frac{i+1}{n+1})}, \quad \hat{Q}_t(1) := +\infty.$$

CRPS for quantile regression

$$\begin{aligned}
 crps(F_t, y) &= \int_{-\infty}^{\infty} \{F_t(x) - \mathbf{1}_{x \geq y}\}^2 dx = \int_{-\infty}^{\infty} \left\{ \sum_{i=1}^n \frac{i}{n} \mathbf{1}_{Q_t^i < x \leq Q_t^{i+1}} - \mathbf{1}_{x \geq y} \right\}^2 dx \\
 &= \int_{-\infty}^{\infty} \sum_{i=1}^n \mathbf{1}_{Q_t^i < x \leq Q_t^{i+1}} \left\{ \frac{i}{n} - \mathbf{1}_{x \geq y} \right\}^2 dx \\
 &= \sum_{i=1}^{n-1} \left\{ \frac{i}{n} - 1 \right\}^2 \int_y^{\infty} \mathbf{1}_{Q_t^i < x \leq Q_t^{i+1}} dx + \sum_{i=1}^n \left\{ \frac{i}{n} \right\}^2 \int_{-\infty}^y \mathbf{1}_{Q_t^i < x \leq Q_t^{i+1}} dx \\
 &= \sum_{i=1}^{n-1} \left\{ \frac{i}{n} - 1 \right\}^2 (Q_t^{i+1} \vee y - Q_t^i \vee y) + \sum_{i=1}^n \left\{ \frac{i}{n} \right\}^2 (Q_t^{i+1} \wedge y - Q_t^i \wedge y)
 \end{aligned}$$

Methods for probabilistic forecasting: parametric approaches

- AR(n) model:

$$X_t = c + a_1 X_{t-1} + \cdots + a_n X_{t-n} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

$\Rightarrow$ for one-step forecasting

$$F_t \sim N(\hat{X}_t, \sigma^2) \quad \text{with} \quad \hat{X}_t = c + a_1 X_{t-1} + \cdots + a_n X_{t-n}$$

- AR(n)-GARCH(1,1) model:

$$X_t = c + a_1 X_{t-1} + \cdots + a_n X_{t-n} + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1)$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

$\Rightarrow$ for one-step forecasting $F_t \sim N(\hat{X}_t, \sigma_t^2)$.

Methods for probabilistic forecasting: nonparametric approaches

- Assume that the unknown variable Y must be predicted from explanatory variables $\mathbf{X} = (X^1, \dots, X^n)$ (e.g., turbine output from meteorological variables).
- The best probabilistic forecast is the conditional density $p(y|\mathbf{x})$ which may be estimated, e.g., by the kernel density estimator

$$\hat{p}(y|\mathbf{x}) = \frac{\hat{p}(y, \mathbf{x})}{\hat{p}(\mathbf{x})} = \frac{\sum_{k=1}^N K_h(y - Y_k) K_h(\mathbf{x} - \mathbf{X}_k)}{\sum_{k=1}^N K_h(\mathbf{x} - \mathbf{X}_k)}, \quad K_h(\cdot) = \frac{1}{h} K(\cdot/h).$$

- If the dimension of the vector of explanatory variables is large, the fully nonparametric estimator suffers from **curse of dimensionality**.

Illustration for wind forecasting

Jupyter notebook for the course

Data file